

Comprehensive Automation Suite

User Manual

Version 1.0

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1. Introduction

The Comprehensive Automation Suite is a Python-based application that brings together multiple independent automation tools into one system. The suite is intended to simplify repetitive computer-based tasks while keeping each automation tool organized as a separate module.

The application provides both Command-Line Interface (CLI) and Graphical User Interface (GUI) interaction. Common services such as configuration, logging, and reporting support the automation modules without becoming part of their individual business logic.

This manual explains the currently available features, the basic operation of the application, and the expected user flow for the File Organizer and Web Scraper.

2. Main Features

The current application provides the following main features:

- File Organizer for automatic file categorization and organization.
- Web Scraper with working CLI functionality for web data extraction.
- Tkinter-based GUI for interacting with the application.
- CLI access for automation tasks without the graphical interface.
- General logging for recording important application events.
- Report generation for automation results, including File Organizer operations.
- Configuration support for application-level settings.

The suite is designed to be expandable, allowing additional automation tools to be integrated without replacing the existing modules.

3. System Requirements

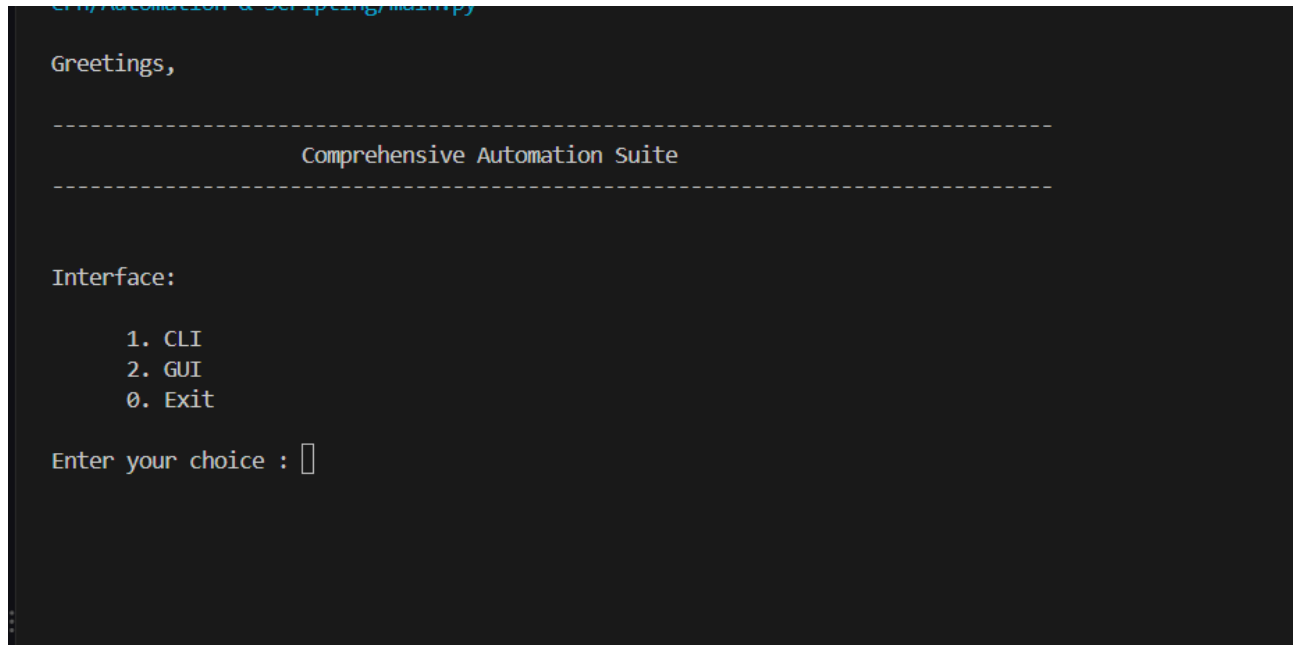
The application requires a compatible Python environment and the packages listed in requirements.txt. The project should be run from the configured project environment so that all required dependencies are available.

The application also requires normal file-system access for file organization and network access when using the Web Scraper.

4. Starting the Application

The project uses `main.py` as the main application entry point. Depending on the intended workflow, the application can be used through the command-line interface or the graphical interface.

The CLI is useful when the user wants direct access to an automation operation. The GUI provides a more visual way to select and operate the available tools.

A screenshot of a terminal window displaying the application's home interface. The text is as follows:

```
erm/automation & scraping/main.py
Greetings,
-----
Comprehensive Automation Suite
-----

Interface:
    1. CLI
    2. GUI
    0. Exit

Enter your choice : 
```

Figure 1 – Application Home Interface

5. Graphical User Interface

The GUI is designed as a single application interface rather than a collection of separate applications. The home interface provides the main area from which available automation tools can be accessed.

The GUI separates presentation and automation responsibilities. The interface collects user input, displays results, provides confirmation controls, and communicates with the corresponding automation module. The actual automation logic remains inside the module.

This structure allows the same automation logic to be used independently through the CLI and through the GUI.

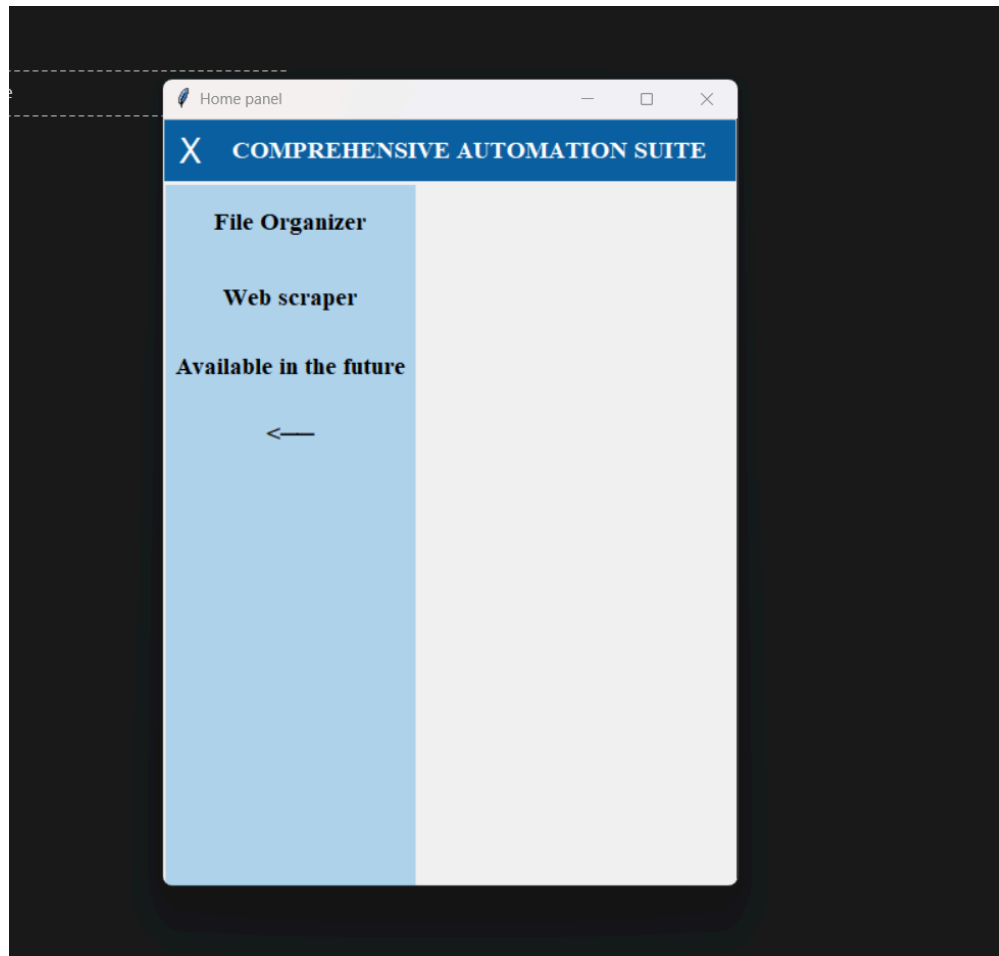


Figure 2 – GUI Home Interface

6. File Organizer

The File Organizer is one of the independent automation tools within the suite. It scans a selected directory, determines suitable categories for files, creates a preview of the planned changes, and moves the files after user confirmation.

The File Organizer currently supports the following categories:

- Image
- Audio
- Video
- Documents
- PDF
- Spreadsheets
- Archives
- Other

Files that do not match the supported categories are handled through the Other category. This gives the organizer a safe fallback instead of leaving an unclassified file without a destination.

6.1 Selecting a Folder

The user begins by selecting the folder that contains the files to be organized. The selected path is validated before the organization process continues.

The selected directory should be a location that the user is authorized to modify. Users should avoid selecting system-critical directories unless they understand the possible effects of moving files from that location.

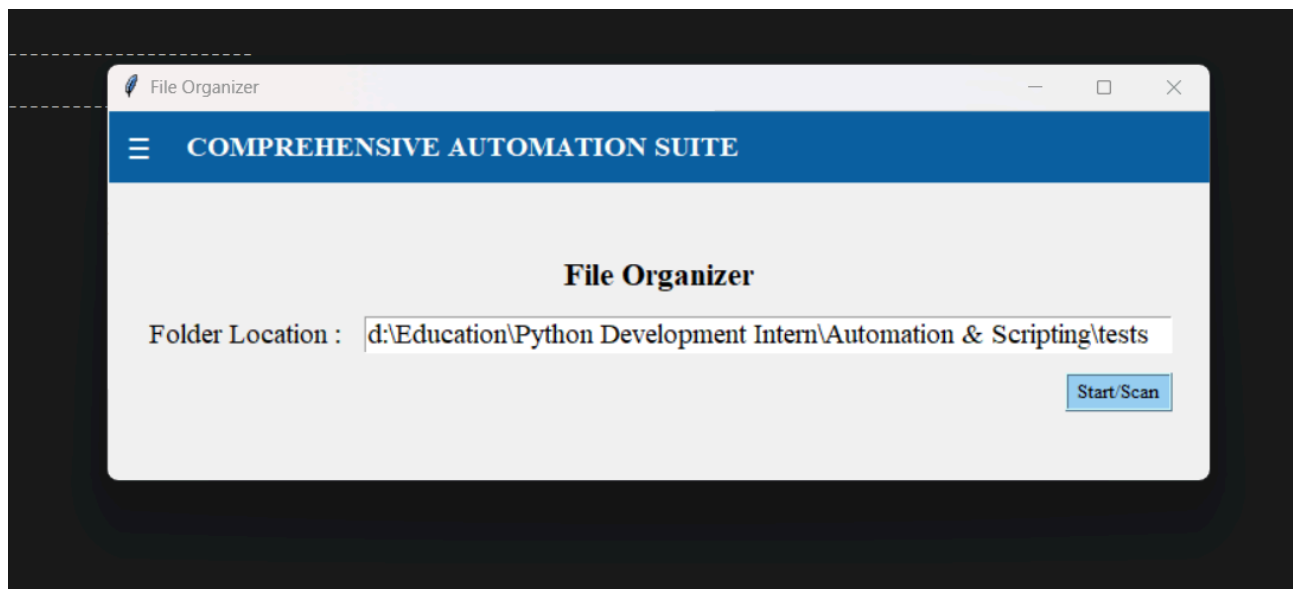


Figure 3 – File Organizer Folder Selection

6.2 Scanning the Directory

After a valid directory is selected, the File Organizer scans the available files. During this stage, the application identifies the files that can be considered for organization and prepares the information needed for categorization.

The scan is kept separate from the final move operation so that the user can review the planned changes before any files are moved.

6.3 File Categorization

Each file is assigned to a category according to its file type. The destination is then determined from that category.

The categorization process is part of the File Organizer module and is not duplicated inside the GUI. The GUI displays the result produced by the automation module.

File → File Type → Category → Destination

6.4 Preview

The File Organizer creates a preview before performing the actual organization. The preview allows the user to see what will happen before confirming the operation.

The preview contains information such as the file name, selected category, and destination. This is especially useful when organizing a directory containing many different file types.

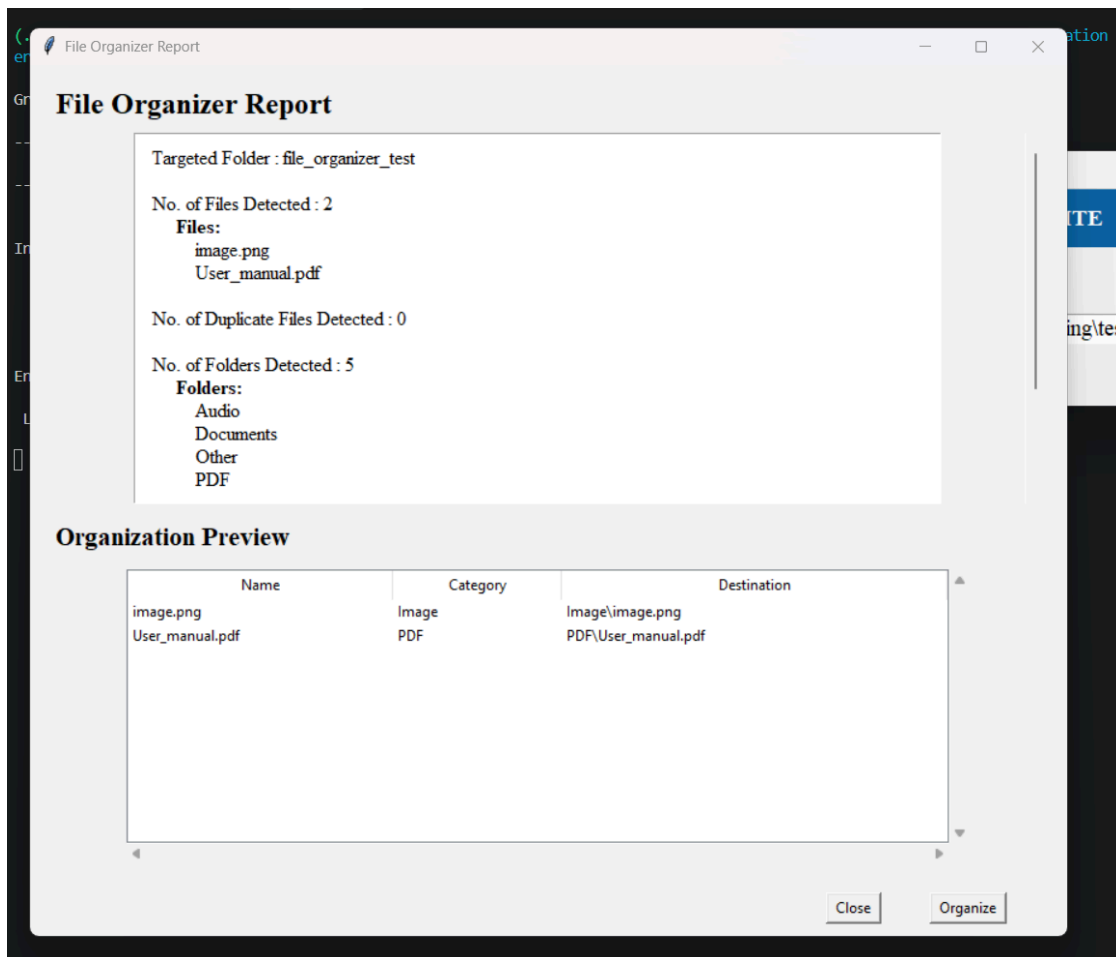


Figure 4 – File Organizer Preview

6.5 Confirmation

After reviewing the preview, the user can either continue with the organization or cancel the operation.

- Confirm / Organize — continues with the file-moving operation.
- Cancel — stops the operation without moving the planned files.

The confirmation stage is an important safety step because file organization changes the location of files on the computer.

6.6 Duplicate Files

The File Organizer includes duplicate filename handling. When a file with the same name already exists at the destination, the application generates a modified filename rather than simply overwriting the existing file.

This helps preserve the existing destination file while allowing the new file to be organized.

6.7 Organization Result

After the move operation is completed, the application provides a result to the user. Important events can also be recorded by the shared logging system.

The File Organizer also has report-generation support so that the result of an organization operation can be preserved in a readable report.

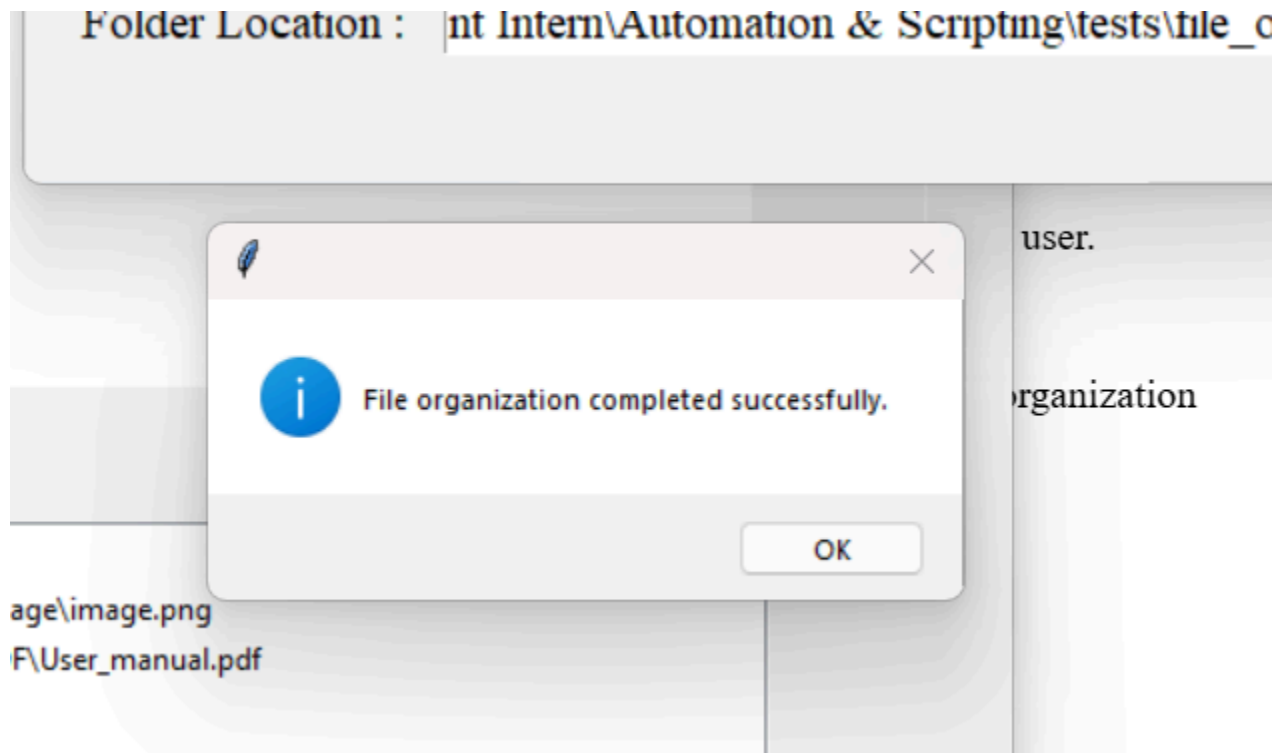


Figure 5 – File Organizer Completion Result

7. Web Scraper

The Web Scraper is another independent automation module within the suite. It is currently functional through the CLI.

The scraper accepts a target URL, performs the required web request, processes the response, and extracts information according to the implemented scraping logic.

The Web Scraper is separate from the File Organizer and does not depend on the File Organizer's operation.

```
ern/automation & scripting/main.py
Greetings,

-----
Comprehensive Automation Suite
-----

Interface:

1. CLI
2. GUI
0. Exit

Enter your choice : 1

Loading the Interface...

Automation tools:

1. File Organizer
2. Web Scraper
0. Exit

Enter your choice : 2

Give url for web scraping : https://www.myntra.com/headphones/boat/boat-rockerz-371-wireless-over-ear-headphones/38208480/buy

What you want to search? : title

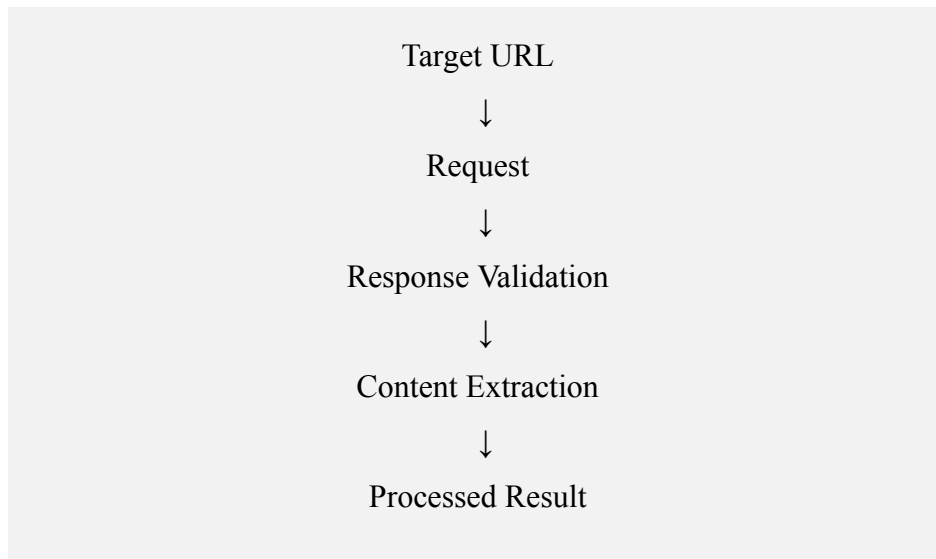
Web page retrieved successfully.
Search result:

1 matches are found.

Buy boAt Rockerz 371 Wireless Over Ear Headphones - Headphones for Unisex 38208480 | Myntra
-----
Comprehensive Automation Suite
```

Figure 6 – Web Scraper CLI

7.1 Basic Scraping Flow



The exact extracted information depends on the target page and the functionality implemented by the scraper.

7.2 Error Handling

Web requests can fail for several reasons, including an invalid URL, connection problems, timeouts, or an unsuccessful server response. The scraper is designed with request handling and error conditions in mind rather than assuming that every request will succeed.

When an operation fails, the user should review the CLI output and available logs for additional information.

7.3 Responsible Web Scraping

The Web Scraper should be used responsibly. Users should collect publicly accessible information and avoid excessive traffic or attempts to bypass access controls.

- Use reasonable request rates.
- Avoid unnecessary repeated requests.
- Respect applicable website rules and terms.
- Do not attempt to bypass authentication or access controls.
- Use scraping only for legitimate and appropriate purposes.

8. CLI Usage

The Command-Line Interface provides direct access to automation operations without requiring the GUI. This is useful when the user wants a lightweight interaction method or wants to operate a tool directly from a terminal.

The current CLI functionality includes the File Organizer and the working Web Scraper. The CLI and GUI are intended to use the same underlying automation modules where applicable.

```
ern/automation & scripting/main.py

Greetings,

-----
Comprehensive Automation Suite
-----

Interface:

    1. CLI
    2. GUI
    0. Exit

Enter your choice : 1

Loading the Interface...

Automation tools:

    1. File Organizer
    2. Web Scraper
    0. Exit

Enter your choice : 0
```

Figure 7 – Command-Line Interface

9. Logging

The suite uses a shared general logging foundation so that modules can record application events in a consistent way. The logging system is intended for application history and troubleshooting rather than replacing the user-facing messages shown by the CLI or GUI.

- Timestamp — when the event occurred.
- Log level — the importance or type of event.
- Module/source — the component that produced the event.
- Message — a description of the event.

The initial logging levels used by the project are INFO, WARNING, and ERROR.

```

183 2026-10-08 22:37:16,259 | INFO | file_organizer | Initializing file scanning....
184 2026-10-08 22:37:16,259 | INFO | file_organizer | File scanning is completed.
185 2026-10-08 22:37:16,259 | INFO | file_organizer | Constructing preview records.
186 2026-10-08 22:37:16,259 | INFO | file_organizer | Preview construction is completed.
187 2026-10-08 22:37:16,259 | INFO | file_organizer | New folders are created to organize.
188 2026-10-08 22:37:30,362 | WARNING | file_organizer | File organization is terminated by user.
189 2026-10-08 22:43:43,832 | INFO | file_organizer | File organizer was started.
190 2026-10-08 22:43:43,832 | INFO | file_organizer | Initializing file scanning....
191 2026-10-08 22:46:47,681 | INFO | file_organizer | File organizer was started.
192 2026-10-08 22:46:47,681 | ERROR | file_organizer | Invalid target path!
193 2026-10-08 22:47:05,222 | INFO | file_organizer | File organizer was started.
194 2026-10-08 22:47:05,222 | INFO | file_organizer | Initializing file scanning....
195 2026-10-08 22:47:05,222 | INFO | file_organizer | File scanning is completed.
196 2026-10-08 22:47:05,222 | INFO | file_organizer | Constructing preview records.
197 2026-10-08 22:47:05,222 | INFO | file_organizer | Preview construction is completed.
198 2026-10-08 22:47:05,224 | INFO | file_organizer | New folders are created to organize.
199 2026-10-08 22:47:21,501 | WARNING | file_organizer | File organization is terminated by user.
200 2026-10-08 22:48:25,887 | INFO | file_organizer | File organizer was started.
201 2026-10-08 22:48:25,887 | INFO | file_organizer | Initializing file scanning....
202 2026-10-08 22:48:25,887 | INFO | file_organizer | File scanning is completed.
203 2026-10-08 22:48:25,887 | INFO | file_organizer | Constructing preview records.
204 2026-10-08 22:48:25,887 | INFO | file_organizer | Preview construction is completed.

```

Figure 8 – Application Log

10. Reports

The suite includes report-generation functionality for automation results. The current project contains dedicated report-generation support for the File Organizer.

Reports provide a more permanent and readable summary of an operation than temporary status messages shown by the application.

```

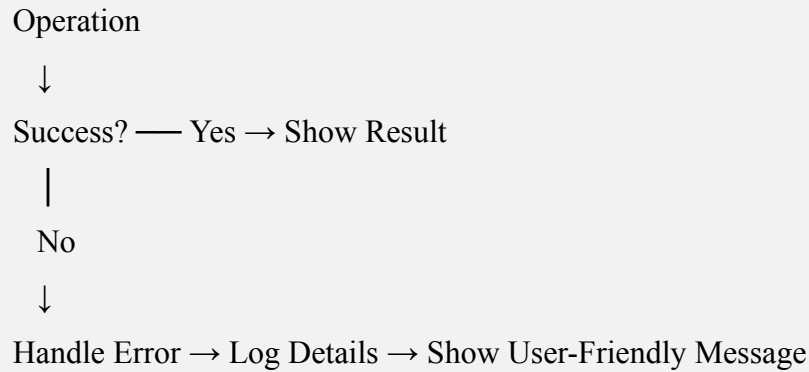
09-Oct-2026_organizer_reports.json X
reports > 09-Oct-2026_organizer_reports.json > ...
1  {
2      "Targeted Folder": "file_organizer_test",
3      "No. of Files Detected": 2,
4      "No. of Duplicate Files Detected": 0,
5      "No. of Folders Detected": 5,
6      "Files": [
7          "image.png",
8          "User_manual.pdf"
9      ],
10     "Folders": [
11         "Audio",
12         "Documents",
13         "Other",
14         "PDF",
15         "Spreadsheets"
16     ],
17     "Preview records": [
18         {
19             "Name": "image.png",
20             "Category": "Image",
21             "Destination": "Image\\image.png"
22         },
23         {
24             "Name": "User_manual.pdf",
25             "Category": "PDF",
26             "Destination": "PDF\\User_manual.pdf"
27         }
28     ],
29     "Message": "File organization completed successfully.",
30     "Time": "03-57-42"
31 }

```

Figure 9 – Generated File Organizer Report

11. Error Handling and User Feedback

The application separates technical logging from user-facing feedback. A technical error can be recorded by the logger while the GUI or CLI provides a simpler message that helps the user understand what action is required.



For example, instead of exposing a raw programming exception, the GUI can provide a clear message asking the user to check the selected directory or operation.

12. Project Folders

The current project is organized into separate folders according to responsibility.

```
configs/  
    general_logger.py  
  
gui/  
    gui_file_organizer.py  
    home_interface.py  
  
modules/  
    file_organizer.py  
    web_scraper.py  
  
reports/  
  
tests/  
    file_organizer_test  
  
utils/  
    organizer_report_generation.py  
  
.gitignore  
config.py  
main.py  
requirements.txt
```

The separation helps keep automation modules, interface components, configuration, reports, tests, and supporting utilities from being mixed together.

13. Safe Usage Recommendations

1. Select the correct directory before starting a File Organizer operation.
2. Review the File Organizer preview before confirming the move.

3. Avoid organizing system-critical directories unless you fully understand the consequences.
4. Review generated reports when you need a record of an operation.
5. Use the Web Scraper only with appropriate publicly accessible content.
6. Check logs when an operation produces an unexpected result.

14. Current Development Status

The Comprehensive Automation Suite is an actively developed modular application. The File Organizer is available through the CLI and has GUI integration. The Web Scraper has working CLI functionality, while its GUI integration is still part of the continuing interface development.

The project also contains the general logging foundation, report-generation support, configuration components, tests, and the GUI application structure.

The suite is designed so that additional automation tools can be added later without changing the purpose of the existing modules.

15. Future Expansion

The architecture allows the suite to grow beyond the currently available tools. Potential expansion areas include additional automation modules, scheduling, system monitoring, email automation, workflow execution, and further GUI integration.

These additions are intended to become separate automation capabilities within the same suite rather than turning the project into one large automation module.

16. Conclusion

The Comprehensive Automation Suite provides a single application environment for multiple independent automation tools. The current implementation demonstrates file organization, web scraping, logging, reporting, CLI operation, and GUI integration.

The modular structure keeps each automation task focused while providing shared services and interfaces around them. This makes the application easier to maintain and gives the project a clear path for future expansion.